复合函数求导

课后作业：习题册 5.4

下周一作业交到第 4 页

03/13/2026



电子科技大学

UNIVERSITY OF ELECTRONIC SCIENCE AND TECHNOLOGY OF CHINA

内容目录 ☞

1 复合函数求导的链式法则
- 链式法则
- 复合函数求导相关例题

2 一阶全微分形式的不变性

3 复合函数的高阶偏导数

链式法则相关注记

- 自变量和因变量由问题本身确定:
  - 公式项数由外层函数是几元函数确定;



链式法则相关注记

- 自变量和因变量由问题本身确定:
  - 公式项数由外层函数是几元函数确定;
  - 能表达的偏导个数由自变量的个数确定.



链式法则相关注记

♠ 自变量和因变量由问题本身确定:

- 公式项数由外层函数是几元函数确定;
- 能表达的偏导个数由自变量的个数确定.

♣ 变量的关系可以按如下方式表达:

$$z = f(u, v), u = u(x, y), v = v(x, y)$$



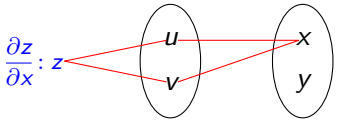
链式法则相关注记

♠ 自变量和因变量由问题本身确定:

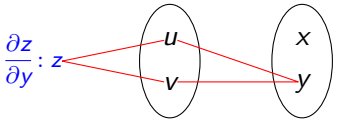
- 公式项数由外层函数是几元函数确定;
- 能表达的偏导个数由自变量的个数确定.

♣ 变量的关系可以按如下方式表达:

$$z = f(u, v), u = u(x, y), v = v(x, y)$$



$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x}.$$



$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y}.$$

非标准情况的链式法则

1).中间变量多于两个时的情况:

设 $z = f(u, v, w), u = u(x, y), v = v(x, y), w = w(x, y)$, 则



非标准情况的链式法则

1).中间变量多于两个时的情况:

设 $z = f(u, v, w), u = u(x, y), v = v(x, y), w = w(x, y)$, 则

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial f}{\partial w}\frac{\partial w}{\partial x},$$

$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y} + \frac{\partial f}{\partial w}\frac{\partial w}{\partial y}.$$



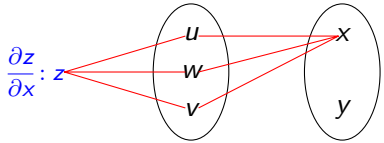
非标准情况的链式法则

1).中间变量多于两个时的情况:

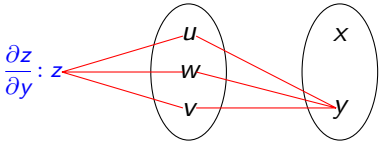
设 $z=f(u,v,w), u=u(x,y), v=v(x,y), w=w(x,y)$, 则

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x}+\frac{\partial f}{\partial w}\frac{\partial w}{\partial x},$$

$$\frac{\partial z}{\partial y}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial y}+\frac{\partial f}{\partial w}\frac{\partial w}{\partial y}.$$



$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x}+\frac{\partial f}{\partial w}\frac{\partial w}{\partial x}.$$



$$\frac{\partial z}{\partial y}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial y}+\frac{\partial f}{\partial w}\frac{\partial w}{\partial y}.$$

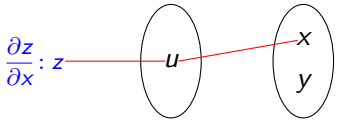
非标准情况的链式法则

2).中间变量只有一个的情况:

设 $z=f(u), u=u(x,y)$, 则

$$\frac{\partial z}{\partial x}=\frac{df}{du}\frac{\partial u}{\partial x}=f'(u)\frac{\partial u}{\partial x},$$

$$\frac{\partial z}{\partial y}=\frac{df}{du}\frac{\partial u}{\partial y}=f'(u)\frac{\partial u}{\partial y}.$$



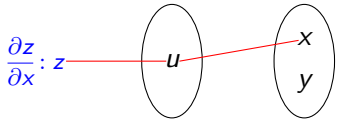
$$\frac{\partial z}{\partial x}=\frac{df}{du}\frac{\partial u}{\partial x}=f'(u)\frac{\partial u}{\partial x}.$$

非标准情况的链式法则

2). 中间变量只有一个的情况:

设 $z=f(u), u=u(x,y)$, 则

$$\frac{\partial z}{\partial x}=\frac{df}{du}\frac{\partial u}{\partial x}=f'(u)\frac{\partial u}{\partial x},$$
$$\frac{\partial z}{\partial y}=\frac{df}{du}\frac{\partial u}{\partial y}=f'(u)\frac{\partial u}{\partial y}.$$

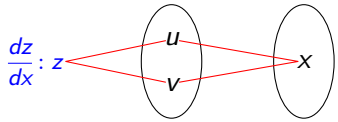


$$\frac{\partial z}{\partial x}=\frac{df}{du}\frac{\partial u}{\partial x}=f'(u)\frac{\partial u}{\partial x}.$$

3). 自变量只有一个时的情况:

设 $z=f(u,v), u=u(x), v=v(x)$, 则 z 是 x 的一元复合函数, 它对 x 的导数称为全导数, 有

$$\frac{dz}{dx}=\frac{\partial f}{\partial u}\frac{du}{dx}+\frac{\partial f}{\partial v}\frac{dv}{dx}.$$



$$\frac{dz}{dx}=\frac{\partial f}{\partial u}\frac{du}{dx}+\frac{\partial f}{\partial v}\frac{dv}{dx}.$$

非标准情况的链式法则

4).混合情况:

$z = f(x, y, t), x = x(s, t), y = y(s, t)$, 则

$$z = f(x(s, t), y(s, t), t).$$

非标准情况的链式法则

4).混合情况:

$z = f(x, y, t), x = x(s, t), y = y(s, t)$, 则

$$z = f(x(s, t), y(s, t), t).$$

从而

$$\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s},$$

$$\frac{\partial z}{\partial t} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial f}{\partial t}.$$

非标准情况的链式法则

4).混合情况:

$z = f(x, y, t), x = x(s, t), y = y(s, t)$, 则

$$z = f(x(s, t), y(s, t), t).$$

从而

$$\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s},$$

$$\frac{\partial z}{\partial t} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial f}{\partial t}.$$

问题:

上式左端的 $\frac{\partial z}{\partial t}$ 和上式右端的 $\frac{\partial f}{\partial t}$ 是否相同？为什么？

链式法则相关例题

例 2

设 $z = uv + \sin t, u = e^t, v = \cos t,$ 求 $\dfrac{dz}{dt}.$

链式法则相关例题

例 2

设 $z = uv + \sin t, u = e^t, v = \cos t$, 求 $\dfrac{dz}{dt}$.

解：由链式法则，直接求导数有

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial u}\cdot\frac{du}{dt} + \frac{\partial z}{\partial v}\cdot\frac{dv}{dt} + \frac{\partial z}{\partial t}\\ &= ve^t - u\sin t + \cos t\\ &= e^t\cos t - e^t\sin t + \cos t\\ &= e^t(\cos t - \sin t) + \cos t.\end{aligned}$$

链式法则相关例题

例 2

设 $z = uv + \sin t, u = e^t, v = \cos t$, 求 $\dfrac{dz}{dt}$.

解: 由链式法则, 直接求导数有

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial u}\cdot\frac{du}{dt} + \frac{\partial z}{\partial v}\cdot\frac{dv}{dt} + \frac{\partial z}{\partial t}\\ &= ve^t - u\sin t + \cos t\\ &= e^t\cos t - e^t\sin t + \cos t\\ &= e^t(\cos t - \sin t) + \cos t.\end{aligned}$$

问题:

先设函数 $z = f(x, x)$, 求 $\dfrac{dz}{dx}$. 再设 $f(x, y) = x^y$, 对 $z = f(x, x)$ 求 $\dfrac{dz}{dx}$.

全微分相关例题

例 3

设 $z = f\left(x^2 y, \dfrac{y}{x}\right)$ 在 $\mathbf{R}^2$ 内具有一阶连续偏导数，求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$.

全微分相关例题

例 3

设 $z = f\left(x^2 y, \frac{y}{x}\right)$ 在 $\mathbf{R}^2$ 内具有一阶连续偏导数，求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

解：记 $u = x^2 y, v = \frac{y}{x}$，则有 $z = f(u, v)$. 由链式法则：

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} = 2xy\frac{\partial f}{\partial u} - \frac{y}{x^2}\frac{\partial f}{\partial v},$$

$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y} = x^2\frac{\partial f}{\partial u} + \frac{1}{x}\frac{\partial f}{\partial v}.$$

链式法则相关例题

例 4

设 f 可微，$z = f(x, e^x)$，求 $\dfrac{dz}{dx}$.

链式法则相关例题

例 4

设 f 可微，$z = f(x, e^x)$，求 $\dfrac{dz}{dx}$.

解：引入中间变量 $y = y(x) = e^x$，则有 $z = f(x, y)$. 从而

$$\frac{dz}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx} = \frac{\partial f}{\partial x} + e^x\frac{\partial f}{\partial y} = f_x + e^x f_y.$$

链式法则相关例题

例 4

设 f 可微，$z = f(x, e^x)$，求 $\dfrac{dz}{dx}$.

解：引入中间变量 $y = y(x) = e^x$，则有 $z = f(x, y)$. 从而

$$\frac{dz}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx} = \frac{\partial f}{\partial x} + e^x\frac{\partial f}{\partial y} = f_x + e^x f_y.$$

例 5

设 f 可微，$z = f\left(x - 2y^2\right)$，求 $\dfrac{\partial z}{\partial x}$, $\dfrac{\partial z}{\partial y}$.



链式法则相关例题

例 4

设 f 可微，$z = f(x, e^x)$，求 $\dfrac{dz}{dx}$.

解：引入中间变量 $y = y(x) = e^x$，则有 $z = f(x, y)$. 从而

$$\frac{dz}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx} = \frac{\partial f}{\partial x} + e^x\frac{\partial f}{\partial y} = f_x + e^x f_y.$$

例 5

设 f 可微，$z = f\left(x - 2y^2\right)$，求 $\dfrac{\partial z}{\partial x}$, $\dfrac{\partial z}{\partial y}$.

解：令 $u = x - 2y^2$，则有 $z = f(u)$. 从而

$$\frac{\partial z}{\partial x} = \frac{df}{du}\frac{\partial u}{\partial x} = f'(u) = f'\left(x - 2y^2\right),$$

$$\frac{\partial z}{\partial y} = \frac{df}{du}\frac{\partial u}{\partial y} = -4yf'(u) = -4yf'\left(x - 2y^2\right).$$

链式法则相关例题

例 6

设 $u = f(x, y)$ 的偏导数表达式为 $\left(\dfrac{\partial u}{\partial x}\right)^2 + \left(\dfrac{\partial u}{\partial y}\right)^2$，求出它在极坐标系下的表达式.

链式法则相关例题

例 6

设 $u = f(x, y)$ 的偏导数表达式为 $\left(\dfrac{\partial u}{\partial x}\right)^2 + \left(\dfrac{\partial u}{\partial y}\right)^2$，求出它在极坐标系下的表达式.

解：直角坐标系与极坐标系的关系为：$\begin{cases} x = r\cos\theta, \\ y = r\sin\theta. \end{cases}$ 即有：

$$u = u(x, y), \quad x = x(r, \theta), \quad y = y(r, \theta),$$

链式法则相关例题

例 6

设 $u = f(x, y)$ 的偏导数表达式为 $\left(\dfrac{\partial u}{\partial x}\right)^2 + \left(\dfrac{\partial u}{\partial y}\right)^2$，求出它在极坐标系下的表达式.

解：直角坐标系与极坐标系的关系为：$\begin{cases} x = r\cos\theta, \\ y = r\sin\theta. \end{cases}$ 即有：

$$u = u(x, y), \quad x = x(r, \theta), \quad y = y(r, \theta),$$

由链式法则有：

$$\begin{cases} \dfrac{\partial u}{\partial r} = \dfrac{\partial u}{\partial x}\dfrac{\partial x}{\partial r} + \dfrac{\partial u}{\partial y}\dfrac{\partial y}{\partial r} = \cos\theta\dfrac{\partial u}{\partial x} + \sin\theta\dfrac{\partial u}{\partial y}, \\ \dfrac{\partial u}{\partial \theta} = \dfrac{\partial u}{\partial x}\dfrac{\partial x}{\partial \theta} + \dfrac{\partial u}{\partial y}\dfrac{\partial y}{\partial \theta} = -r\sin\theta\dfrac{\partial u}{\partial x} + r\cos\theta\dfrac{\partial u}{\partial y}. \end{cases}$$

链式法则相关例题

例 6

设 $u=f(x,y)$ 的偏导数表达式为 $\left(\dfrac{\partial u}{\partial x}\right)^2+\left(\dfrac{\partial u}{\partial y}\right)^2$，求出它在极坐标系下的表达式.

解：直角坐标系与极坐标系的关系为：$\begin{cases} x=r\cos\theta, \\ y=r\sin\theta. \end{cases}$ 即有：

$$u=u(x,y),\quad x=x(r,\theta),\quad y=y(r,\theta),$$

由链式法则有：

$$\begin{cases} \dfrac{\partial u}{\partial r}=\dfrac{\partial u}{\partial x}\dfrac{\partial x}{\partial r}+\dfrac{\partial u}{\partial y}\dfrac{\partial y}{\partial r}=\cos\theta\dfrac{\partial u}{\partial x}+\sin\theta\dfrac{\partial u}{\partial y}, \\ \dfrac{\partial u}{\partial \theta}=\dfrac{\partial u}{\partial x}\dfrac{\partial x}{\partial \theta}+\dfrac{\partial u}{\partial y}\dfrac{\partial y}{\partial \theta}=-r\sin\theta\dfrac{\partial u}{\partial x}+r\cos\theta\dfrac{\partial u}{\partial y}. \end{cases}$$

从而 $\begin{cases} \dfrac{\partial u}{\partial x}=\cos\theta\dfrac{\partial u}{\partial r}-\dfrac{1}{r}\sin\theta\dfrac{\partial u}{\partial \theta} \\ \dfrac{\partial u}{\partial y}=\sin\theta\dfrac{\partial u}{\partial r}+\dfrac{1}{r}\cos\theta\dfrac{\partial u}{\partial \theta} \end{cases}$，因此

$$\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2=\left(\frac{\partial u}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2.$$

链式法则相关例题

例 7

设 $z = f(x, y, u) = xy + xF(u)$，其中 $F(u)$ 可微，并且 $u = \dfrac{y}{x}$，试求：$x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y}$ 的值.

链式法则相关例题

例 7

设 $z = f(x, y, u) = xy + xF(u)$，其中 $F(u)$ 可微，并且 $u = \dfrac{y}{x}$，试求：$x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y}$ 的值.

解： 把函数看作 $z = f(x, y, u), u = \dfrac{y}{x}$，故由复合函数求导法则

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} = y + F(u) + xF'(u)\left(-\frac{y}{x^2}\right) = y + F(u) - \frac{y}{x}F'(u),$$

$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} + \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} = x + xF'(u)\frac{1}{x} = x + F'(u).$$

因此，

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = x\left(y + F(u) - \frac{y}{x}F'(u)\right) + y\left(x + F'(u)\right) = 2xy + xF(u).$$

链式法则相关总结

♠ 多元复合函数求偏导步聚公式求导法:

1. 根据函数方程和问题判定因变量、自变量和中间变量，理清复合关系与过程;
2. 画出变量层次图;
3. 根据多元复合函数求偏导的链式法则写出计算公式:

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x},\quad \frac{\partial z}{\partial y}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial y}.$$

链式法则相关总结

♠ 多元复合函数求偏导步聚公式求导法:

1. 根据函数方程和问题判定因变量、自变量和中间变量，理清复合关系与过程;
2. 画出变量层次图;
3. 根据多元复合函数求偏导的链式法则写出计算公式:

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x},\quad \frac{\partial z}{\partial y}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial y}.$$

♣ 多元函数求偏导方法步骤直接求导法:

1. 根据函数方程和问题判定因变量、自变量和中间变量（关于谁的函数）;
2. 根据求导对象选择求导的四则运算法则、一元复合函数求导的链式法则或多元复合函数求导的链式法则.

链式法则相关练习

练习 1

设 $z=f(u,v), u=\dfrac{x}{y}, v=x^2+y^2$，则 $dz=$（　　）

$(A).\left(-\dfrac{1}{y}\dfrac{\partial f}{\partial u}+2x\dfrac{\partial f}{\partial v}\right)dx+\left(-\dfrac{1}{y^2}\dfrac{\partial f}{\partial u}+2y\dfrac{\partial f}{\partial v}\right)dy;$

$(B).\left(-\dfrac{1}{y^2}\dfrac{\partial f}{\partial u}+2y\dfrac{\partial f}{\partial v}\right)dx+\left(-\dfrac{1}{y}\dfrac{\partial f}{\partial u}+2x\dfrac{\partial f}{\partial v}\right)dy;$

$(C).\left(\dfrac{1}{y}\dfrac{\partial f}{\partial u}+2x\dfrac{\partial f}{\partial v}\right)dx+\left(-\dfrac{x}{y^2}\dfrac{\partial f}{\partial u}+2y\dfrac{\partial f}{\partial v}\right)dy;$

$(D).\left(-\dfrac{x}{y^2}\dfrac{\partial f}{\partial u}+2y\dfrac{\partial f}{\partial v}\right)dx+\left(\dfrac{1}{y}\dfrac{\partial f}{\partial u}+2x\dfrac{\partial f}{\partial v}\right)dy.$

练习 1 解答

解：直接计算有：

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x}=\frac{1}{y}\frac{\partial f}{\partial u}+2x\frac{\partial f}{\partial v},$$
$$\frac{\partial z}{\partial y}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial y}=-\frac{x}{y^2}\frac{\partial f}{\partial u}+2y\frac{\partial f}{\partial v}.$$

所以

$$dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy=\left(\frac{1}{y}\frac{\partial f}{\partial u}+2x\frac{\partial f}{\partial v}\right)dx+\left(-\frac{x}{y^2}\frac{\partial f}{\partial u}+2y\frac{\partial f}{\partial v}\right)dy.$$

故选择 (C).

全微分相关练习

练习 2

若函数 $z = x^n f\left(\frac{y}{x^2}\right)$ 和 $f(u)$ 均可微，则 $x\frac{\partial z}{\partial x} + 2y\frac{\partial z}{\partial y} = (\quad)$

$(A).(n-1)z \qquad (B).nz \qquad (C).(n+1)z \qquad (D).nx^{n-1}z$

全微分相关练习

练习 2

若函数 $z=x^n f\left(\dfrac{y}{x^2}\right)$ 和 $f(u)$ 均可微，则 $x\dfrac{\partial z}{\partial x}+2y\dfrac{\partial z}{\partial y}=(\quad)$

$(A).(n-1)z \qquad (B).nz \qquad (C).(n+1)z \qquad (D).nx^{n-1}z$

解： 利用求导的乘法法则：

$$\frac{\partial z}{\partial x}=nx^{n-1}f\left(\frac{y}{x^2}\right)+x^n f'\left(\frac{y}{x^2}\right)\cdot\left(-\frac{2y}{x^3}\right)=nx^{n-1}f-2x^{n-3}yf',$$

$$\frac{\partial z}{\partial y}=x^n f'\left(\frac{y}{x^2}\right)\cdot\left(\frac{1}{x^2}\right)=x^{n-2}f'.$$

因此，

$$x\frac{\partial z}{\partial x}+2y\frac{\partial z}{\partial y}=nx^n f-2x^{n-2}yf'+2x^{n-2}yf'=nx^n f=nz.$$

故选择 (B).

内容目录

1 复合函数求导的链式法则

2 一阶全微分形式的不变性

3 复合函数的高阶偏导数

一阶全微分形式的不变性

设 $z = f(x, y)$ 可微，如果 x, y 都是自变量，则

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy.$$

一阶全微分形式的不变性

设 $z = f(x, y)$ 可微，如果 x, y 都是自变量，则

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy.$$

若 x, y 不是自变量而是中间变量 $x = x(u, v), y = y(u, v)$，由定义：

$$dz = \frac{\partial z}{\partial u}du + \frac{\partial z}{\partial v}dv.$$

一阶全微分形式的不变性

设 $z = f(x, y)$ 可微，如果 x, y 都是自变量，则

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy.$$

若 x, y 不是自变量而是中间变量 $x = x(u, v), y = y(u, v)$，由定义：

$$dz = \frac{\partial z}{\partial u}du + \frac{\partial z}{\partial v}dv.$$

其中，$\dfrac{\partial z}{\partial u} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}$，$\dfrac{\partial z}{\partial v} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial v} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial v}$.



一阶全微分形式的不变性

设 $z=f(x,y)$ 可微，如果 x,y 都是自变量，则

$$dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy.$$

若 x,y 不是自变量而是中间变量 $x=x(u,v),y=y(u,v)$，由定义：

$$dz=\frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv.$$

其中，$\dfrac{\partial z}{\partial u}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}$，$\dfrac{\partial z}{\partial v}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial v}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial v}$. 因此，

$$dz=\left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial u}\right)du+\left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial v}\right)dv$$



一阶全微分形式的不变性

设 $z=f(x,y)$ 可微，如果 x,y 都是自变量，则

$$dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy.$$

若 x,y 不是自变量而是中间变量 $x=x(u,v),y=y(u,v)$，由定义：

$$dz=\frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv.$$

其中，$\dfrac{\partial z}{\partial u}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}$，$\dfrac{\partial z}{\partial v}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial v}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial v}$. 因此，

$$\begin{aligned}dz&=\left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial u}\right)du+\left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial v}\right)dv\\&=\frac{\partial z}{\partial x}\left(\frac{\partial x}{\partial u}du+\frac{\partial x}{\partial v}dv\right)+\frac{\partial z}{\partial y}\left(\frac{\partial y}{\partial u}du+\frac{\partial y}{\partial v}dv\right)=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy.\end{aligned}$$



一阶全微分形式的不变性

设 $z=f(x,y)$ 可微，如果 x,y 都是自变量，则

$$dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy.$$

若 x,y 不是自变量而是中间变量 $x=x(u,v),y=y(u,v)$，由定义：

$$dz=\frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv.$$

其中，$\dfrac{\partial z}{\partial u}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}$，$\dfrac{\partial z}{\partial v}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial v}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial v}$. 因此，

$$\begin{aligned}dz&=\left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial u}\right)du+\left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial v}\right)dv\\&=\frac{\partial z}{\partial x}\left(\frac{\partial x}{\partial u}du+\frac{\partial x}{\partial v}dv\right)+\frac{\partial z}{\partial y}\left(\frac{\partial y}{\partial u}du+\frac{\partial y}{\partial v}dv\right)=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy.\end{aligned}$$

由此可见：不论 x,y 是自变量还是中间变量，都有

$$dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy.$$

这一性质称为一阶全微分形式的不变性.

全微分运算法则

全微分运算法则:

- $\mathrm{d}(u \pm v) = \mathrm{d}u \pm \mathrm{d}v$;
- $\mathrm{d}(uv) = v\mathrm{d}u + u\mathrm{d}v$;
- $\mathrm{d}\left(\dfrac{u}{v}\right) = \dfrac{v\mathrm{d}u - u\mathrm{d}v}{v^2} \quad (v \neq 0)$;
- $\mathrm{d}f(u) = f'(u)\mathrm{d}u$, 其中 u 是自变量或中间变量;
- $\mathrm{d}f(u,v) = \dfrac{\partial f}{\partial u}\mathrm{d}u + \dfrac{\partial f}{\partial v}\mathrm{d}v$, 其中 u, v 是自变量或中间变量.



链式法则相关例题

例 8

已知 $e^{-xy} - 2z + e^z = 0$，求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$ 和 dz.

链式法则相关例题

例 8

已知 $e^{-xy}-2z+e^z=0$，求 $\dfrac{\partial z}{\partial x},\dfrac{\partial z}{\partial y}$ 和 dz.

解：由于 $d(e^{-xy}-2z+e^z)=0$，故

$$
\begin{aligned}
& e^{-xy}d(-xy)-2dz+e^z dz=0,\\
\Longrightarrow\quad & (e^z-2)\,dz=e^{-xy}(xdy+ydx),\\
\Longrightarrow\quad & dz=\frac{ye^{-xy}}{(e^z-2)}dx+\frac{xe^{-xy}}{(e^z-2)}dy.
\end{aligned}
$$

因此，

$$
\frac{\partial z}{\partial x}=\frac{ye^{-xy}}{e^z-2},\quad \frac{\partial z}{\partial y}=\frac{xe^{-xy}}{e^z-2}.
$$

链式法则相关例题

例 8

已知 $e^{-xy}-2z+e^z=0$，求 $\dfrac{\partial z}{\partial x},\dfrac{\partial z}{\partial y}$ 和 dz.

解：由于 $d(e^{-xy}-2z+e^z)=0$，故

$$\begin{aligned} & e^{-xy}d(-xy)-2dz+e^z dz=0, \\ \Longrightarrow\quad & (e^z-2)\,dz=e^{-xy}(xdy+ydx), \\ \Longrightarrow\quad & dz=\frac{ye^{-xy}}{(e^z-2)}dx+\frac{xe^{-xy}}{(e^z-2)}dy. \end{aligned}$$

因此，

$$\frac{\partial z}{\partial x}=\frac{ye^{-xy}}{e^z-2},\quad \frac{\partial z}{\partial y}=\frac{xe^{-xy}}{e^z-2}.$$

微分特点：不需要区分因变量、自变量与中间变量，按微分法则（对微分对象使用）微分到需要的变量为止.

链式法则相关例题

例 9

由微分形式不变性，求函数 $z = \left(x^2 + y^2\right)^{xy}$ 全微分 dz 和偏导数 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$.

链式法则相关例题

例 9

由微分形式不变性，求函数 $z=\left(x^2+y^2\right)^{xy}$ 全微分 dz 和偏导数 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$.

解： 设 $u=x^2+y^2, v=xy$，则 $z=u^v$. 由一阶微分形式的不变性：

$$dz=\frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv=vu^{v-1}du+u^v\ln u dv=vu^{v-1}d\left(x^2+y^2\right)+u^v\ln u d(xy)$$

链式法则相关例题

例 9

由微分形式不变性，求函数 $z=\left(x^2+y^2\right)^{xy}$ 全微分 dz 和偏导数 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$.

解： 设 $u=x^2+y^2, v=xy$，则 $z=u^v$. 由一阶微分形式的不变性:

$$
\begin{aligned}
dz=&\frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv=vu^{v-1}du+u^v\ln u dv=vu^{v-1}d\left(x^2+y^2\right)+u^v\ln u d(xy)\\
=&vu^{v-1}\left[d\left(x^2\right)+d\left(y^2\right)\right]+u^v\ln u(xdy+ydx)\\
=&vu^{v-1}[2xdx+2ydy]+u^v\ln u(xdy+ydx)\\
=&\left[xy\left(x^2+y^2\right)^{xy-1}\cdot 2x+y\left(x^2+y^2\right)^{xy}\ln\left(x^2+y^2\right)\right]dx\\
&+\left[xy\left(x^2+y^2\right)^{xy-1}\cdot 2y+x\left(x^2+y^2\right)^{xy}\ln\left(x^2+y^2\right)\right]dy.
\end{aligned}
$$

链式法则相关例题

例 9

由微分形式不变性，求函数 $z=\left(x^2+y^2\right)^{xy}$ 全微分 dz 和偏导数 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$.

解：设 $u=x^2+y^2, v=xy$，则 $z=u^v$. 由一阶微分形式的不变性：

$$
\begin{aligned}
dz=&\frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv=vu^{v-1}du+u^v\ln u dv=vu^{v-1}d\left(x^2+y^2\right)+u^v\ln u d(xy)\\
=&vu^{v-1}\left[d\left(x^2\right)+d\left(y^2\right)\right]+u^v\ln u(xdy+ydx)\\
=&vu^{v-1}[2xdx+2ydy]+u^v\ln u(xdy+ydx)\\
=&\left[xy\left(x^2+y^2\right)^{xy-1}\cdot 2x+y\left(x^2+y^2\right)^{xy}\ln\left(x^2+y^2\right)\right]dx\\
&+\left[xy\left(x^2+y^2\right)^{xy-1}\cdot 2y+x\left(x^2+y^2\right)^{xy}\ln\left(x^2+y^2\right)\right]dy.
\end{aligned}
$$

因此我们有

$$
\begin{aligned}
\frac{\partial z}{\partial x}&=xy\left(x^2+y^2\right)^{xy-1}\cdot 2x+y\left(x^2+y^2\right)^{xy}\ln\left(x^2+y^2\right)\\
\frac{\partial z}{\partial y}&=xy\left(x^2+y^2\right)^{xy-1}\cdot 2y+x\left(x^2+y^2\right)^{xy}\ln\left(x^2+y^2\right).
\end{aligned}
$$

全微分相关练习

练习 3

若 $z = x^n f\left(\frac{y}{x^2}\right)$, $f(u)$ 可微，利用微分运算计算 $dz =$ (　　)

A. $\left[nx^{n-1} f\left(\frac{y}{x^2}\right) - 2x^{n-2} y f'\left(\frac{y}{x^2}\right)\right] dx + x^{n-1} f'\left(\frac{y}{x^2}\right) dy$;

B. $x^{n-2} f'\left(\frac{y}{x^2}\right) dx + \left[nx^{n-1} f\left(\frac{y}{x^2}\right) - 2x^{n-3} y f'\left(\frac{y}{x^2}\right)\right] dy$;

C. $x^{n-1} f'\left(\frac{y}{x^2}\right) dx + \left[nx^{n-1} f\left(\frac{y}{x^2}\right) - 2x^{n-2} y f'\left(\frac{y}{x^2}\right)\right] dy$;

D. $\left[nx^{n-1} f\left(\frac{y}{x^2}\right) - 2x^{n-3} y f'\left(\frac{y}{x^2}\right)\right] dx + x^{n-2} f'\left(\frac{y}{x^2}\right) dy$.

练习 3 解答

解: 直接计算有

$$\begin{aligned} dz &= d\left[x^n f\left(\frac{y}{x^2}\right)\right] = f\left(\frac{y}{x^2}\right) dx^n + x^n d\left[f\left(\frac{y}{x^2}\right)\right] \\ &= nx^{n-1} f\left(\frac{y}{x^2}\right) dx + x^n f'\left(\frac{y}{x^2}\right) d\left(\frac{y}{x^2}\right) \\ &= nx^{n-1} f\left(\frac{y}{x^2}\right) dx + x^n f'\left(\frac{y}{x^2}\right) \frac{x^2 dy - y dx^2}{x^4} \\ &= nx^{n-1} f\left(\frac{y}{x^2}\right) dx + x^n f'\left(\frac{y}{x^2}\right) \frac{x^2 dy - 2xy dx}{x^4} \\ &= nx^{n-1} f\left(\frac{y}{x^2}\right) dx + x^{n-3} f'\left(\frac{y}{x^2}\right) (x dy - 2y dx) \\ &= \left[nx^{n-1} f\left(\frac{y}{x^2}\right) - 2x^{n-3} y f'\left(\frac{y}{x^2}\right)\right] dx + x^{n-2} f'\left(\frac{y}{x^2}\right) dy \end{aligned}$$

故选择 (D).

内容目录

1 复合函数求导的链式法则

2 一阶全微分形式的不变性

3 复合函数的高阶偏导数

复合函数的高阶偏导数

复合函数的高阶偏导数有一定难度，下面以例题形式分析和研究.

例 10

设 $z = f\left(x^2y, \frac{y}{x}\right)$，$f$ 具有二阶连续偏导数，求 $\frac{\partial z}{\partial x}, \frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}$.

复合函数的高阶偏导数

复合函数的高阶偏导数有一定难度，下面以例题形式分析和研究.

例 10

设 $z = f\left(x^2y, \dfrac{y}{x}\right)$，$f$ 具有二阶连续偏导数，求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial^2 z}{\partial x^2}, \dfrac{\partial^2 z}{\partial x \partial y}$.

解：设 $u = x^2y$，$v = \dfrac{y}{x}$，则 $z = f(u, v)$，故

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} = 2xy\frac{\partial f}{\partial u} - \frac{y}{x^2}\frac{\partial f}{\partial v} = 2xy \cdot f_u(u, v) - \frac{y}{x^2} \cdot f_v(u, v).$$

其中 f_u 和 f_v 是关于 u, v 的函数，记 $\dfrac{\partial f}{\partial u} = f_u(u, v), \dfrac{\partial f}{\partial v} = f_v(u, v)$.

复合函数的高阶偏导数

复合函数的高阶偏导数有一定难度，下面以例题形式分析和研究.

例 10

设 $z=f\left(x^2y,\dfrac{y}{x}\right)$，$f$ 具有二阶连续偏导数，求 $\dfrac{\partial z}{\partial x},\dfrac{\partial^2 z}{\partial x^2},\dfrac{\partial^2 z}{\partial x\partial y}$.

解：设 $u=x^2y$，$v=\dfrac{y}{x}$，则 $z=f(u,v)$，故

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x}=2xy\frac{\partial f}{\partial u}-\frac{y}{x^2}\frac{\partial f}{\partial v}=2xy\cdot f_u(u,v)-\frac{y}{x^2}\cdot f_v(u,v).$$

其中 f_u 和 f_v 是关于 u,v 的函数，记 $\dfrac{\partial f}{\partial u}=f_u(u,v),\dfrac{\partial f}{\partial v}=f_v(u,v)$. 对于 f_u 和 f_v 使用链式法则得:

$$\begin{aligned}\frac{\partial^2 z}{\partial x^2}=&2y\frac{\partial f}{\partial u}+2xy\frac{\partial f_u}{\partial x}+\frac{2y}{x^3}\frac{\partial f}{\partial v}-\frac{y}{x^2}\frac{\partial f_v}{\partial x}\\=&2y\frac{\partial f}{\partial u}+2xy\left(\frac{\partial^2 f}{\partial u^2}\frac{\partial u}{\partial x}+\frac{\partial^2 f}{\partial u\partial v}\frac{\partial v}{\partial x}\right)+\frac{2y}{x^3}\frac{\partial f}{\partial v}-\frac{y}{x^2}\left(\frac{\partial^2 f}{\partial v\partial u}\frac{\partial u}{\partial x}+\frac{\partial^2 f}{\partial v^2}\frac{\partial v}{\partial x}\right).\end{aligned}$$

例 10 解答

因为 f 具有二阶连续偏导数，所以 $\dfrac{\partial^2 f}{\partial u \partial v} = \dfrac{\partial^2 f}{\partial v \partial u}$. 故

$$\frac{\partial^2 z}{\partial x^2} = 4x^2y^2\frac{\partial^2 f}{\partial u^2} - \frac{4y^2}{x}\frac{\partial^2 f}{\partial u \partial v} + \frac{y^2}{x^4}\frac{\partial^2 f}{\partial v^2} + 2y\frac{\partial f}{\partial u} + \frac{2y}{x^3}\frac{\partial f}{\partial v}.$$

例 10 解答

因为 f 具有二阶连续偏导数，所以 $\dfrac{\partial^2 f}{\partial u \partial v} = \dfrac{\partial^2 f}{\partial v \partial u}$. 故

$$\frac{\partial^2 z}{\partial x^2} = 4x^2y^2\frac{\partial^2 f}{\partial u^2} - \frac{4y^2}{x}\frac{\partial^2 f}{\partial u \partial v} + \frac{y^2}{x^4}\frac{\partial^2 f}{\partial v^2} + 2y\frac{\partial f}{\partial u} + \frac{2y}{x^3}\frac{\partial f}{\partial v}.$$

$$\begin{aligned}
\frac{\partial^2 z}{\partial x \partial y} =& 2x\frac{\partial f}{\partial u} + 2xy\frac{\partial f_u}{\partial y} - \frac{1}{x^2}\frac{\partial f}{\partial v} - \frac{y}{x^2}\frac{\partial f_v}{\partial y} \\
=& 2x\frac{\partial f}{\partial u} + 2xy\left(\frac{\partial^2 f}{\partial u^2}\frac{\partial u}{\partial y} + \frac{\partial^2 f}{\partial u \partial v}\frac{\partial v}{\partial y}\right) \\
& - \frac{1}{x^2}\frac{\partial f}{\partial v} - \frac{y}{x^2}\left(\frac{\partial^2 f}{\partial v \partial u}\frac{\partial u}{\partial y} + \frac{\partial^2 f}{\partial v^2}\frac{\partial v}{\partial y}\right) \\
=& 2x^3y\frac{\partial^2 f}{\partial u^2} + y\frac{\partial^2 f}{\partial u \partial v} - \frac{y}{x^3}\frac{\partial^2 f}{\partial v^2} + 2x\frac{\partial f}{\partial u} - \frac{1}{x^2}\frac{\partial f}{\partial v}.
\end{aligned}$$

高阶导数的记号

为了数学简便，引进符号:

- $f_1 = \dfrac{\partial f}{\partial u} = f_u$，表示函数 f 对第一个中间变量 u 求偏导数;



高阶导数的记号

为了数学简便，引进符号:

- $f_1 = \dfrac{\partial f}{\partial u} = f_u$，表示函数 f 对第一个中间变量 u 求偏导数;
- $f_2 = \dfrac{\partial f}{\partial v} = f_v$，表示函数 f 对第二个中间变量 v 求偏导数;

高阶导数的记号

为了数学简便，引进符号:

- $f_1 = \dfrac{\partial f}{\partial u} = f_u$，表示函数 f 对第一个中间变量 u 求偏导数;
- $f_2 = \dfrac{\partial f}{\partial v} = f_v$，表示函数 f 对第二个中间变量 v 求偏导数;
- $f_{12} = \dfrac{\partial^2 f}{\partial u \partial v} = f_{uv}$，表示函数 f 先对 u 再对 v 求偏导数，以此类推.



高阶导数的记号

为了数学简便，引进符号:

- $f_1 = \dfrac{\partial f}{\partial u} = f_u$，表示函数 f 对第一个中间变量 u 求偏导数;
- $f_2 = \dfrac{\partial f}{\partial v} = f_v$，表示函数 f 对第二个中间变量 v 求偏导数;
- $f_{12} = \dfrac{\partial^2 f}{\partial u \partial v} = f_{uv}$，表示函数 f 先对 u 再对 v 求偏导数，以此类推.

则例 10 结果可记为:

$$
\begin{aligned}
\frac{\partial z}{\partial x} &= 2xy\frac{\partial f}{\partial u} - \frac{y}{x^2}\frac{\partial f}{\partial v} = 2xyf_1 - \frac{y}{x^2}f_2, \\
\frac{\partial^2 z}{\partial x^2} &= 4x^2y^2\frac{\partial^2 f}{\partial u^2} - \frac{4y^2}{x}\frac{\partial^2 f}{\partial u \partial v} + \frac{y^2}{x^4}\frac{\partial^2 f}{\partial v^2} + 2y\frac{\partial f}{\partial u} + \frac{2y}{x^3}\frac{\partial f}{\partial v} \\
&= 4x^2y^2f_{11} - \frac{4y^2}{x}f_{12} + \frac{y^2}{x^4}f_{22} + 2yf_1 + \frac{2y}{x^3}f_2, \\
\frac{\partial^2 z}{\partial x \partial y} &= 2x^3y\frac{\partial^2 f}{\partial u^2} + y\frac{\partial^2 f}{\partial u \partial v} - \frac{y}{x^3}\frac{\partial^2 f}{\partial v^2} + 2x\frac{\partial f}{\partial u} - \frac{1}{x^2}\frac{\partial f}{\partial v} \\
&= 2x^3yf_{11} + yf_{12} - \frac{y}{x^3}f_{22} + 2xf_1 - \frac{1}{x^2}f_2.
\end{aligned}
$$